

Foundation (Jalapeno)

- Q1.** A basketball player's score y after x games is given by $y = 2x + 5$. If the score is shifted down by 3 points, the new equation is
(A) $y = 2x + 2$
(B) $y = 2x - 3$
(C) $y = 2x + 8$
(D) $y = x + 5$
(E) $y = 3x + 2$
- Q2.** The distance d a gymnast travels after t seconds is given by $d = 5t$. If the gymnast starts 2 seconds late, the new equation is
(A) $d = 5t - 10$
(B) $d = 5(t - 2)$
(C) $d = 5t + 2$
(D) $d = t + 5$
(E) $d = 10t$
- Q3.** A tennis player's serve speed y after x practices is given by $y = 10x$. If the speed is reflected over the x-axis, the new equation is
(A) $y = -10x$
(B) $y = 10x$
(C) $y = 5x$
(D) $y = x + 10$
(E) $y = 20x$
- Q4.** The height h of a hurdle after x adjustments is given by $h = 3x$. If the hurdle is stretched vertically by a factor of 2, the new equation is
(A) $h = 3x$
(B) $h = 6x$
(C) $h = 2x$
(D) $h = x + 3$
(E) $h = 9x$
- Q5.** A cyclist's speed s after t hours is given by $s = 15t$. If the speed is compressed horizontally by a factor of 2, the new equation is
(A) $s = 15t$
(B) $s = 30t$
(C) $s = 7.5t$
(D) $s = t + 15$
(E) $s = 60t$
- Q6.** The score y of a swimmer after x laps is given by $y = x^2 + 2$. If the score is shifted left by 1 lap, the new equation is
(A) $y = (x + 1)^2 + 2$
(B) $y = (x - 1)^2 + 2$
(C) $y = x^2 - 2$
(D) $y = x^2 + 1$
(E) $y = x^2 + 4$
- Q7.** A runner's distance d after t minutes is given by $d = 8t$. If the runner's distance is reflected over the y-axis, the new equation is
(A) $d = -8t$
(B) $d = 8t$
(C) $d = 4t$
(D) $d = t + 8$
(E) $d = 16t$
- Q8.** The time t it takes for a skier to finish a course is given by $t = 2x + 1$, where x is the number of practices. If the time is stretched vertically by a factor of 3, the new equation is
(A) $t = 2x + 1$
(B) $t = 6x + 1$
(C) $t = 3x + 1$
(D) $t = x + 2$
(E) $t = 9x + 3$

Open Ended Questions (FRQ)

- Q9.** Graph the function $y = 2x^2 + 1$ and its reflection over the x-axis. Label the parent function and the transformed function.
- Q10.** Compare the graphs of $y = x^2$ and $y = 3(x - 2)^2 + 1$. Describe the transformations that occur and provide the equation of the transformed function.

Chef's Tips

- Tip 1: Use graph paper to help students visualize the transformations.
- Tip 2: Encourage students to use mathematical notation and terminology when describing the transformations.
- Tip 3: Consider using real-world examples, such as sports and fitness, to illustrate the transformations and make the concepts more relatable.